

MKTS_Module_model_summary.txt

TBD2 pipeline output - rendered to PDF

NOTE ON THE SECTION BELOW: 'Overall Risk', the 'Risk Scenarios' ranking, and 'Follow-up Actions' are computed from randomly simulated stress-test scenarios (seeded for reproducibility), not from statistical evidence measured in this client's actual data. Treat them as illustrative stress-test projections, not as data-driven findings. (Model Evaluation Metrics further below ARE fit on the real data and are not affected by this note.)

Overall Risk: 0.38030394934865536

Risk Scenarios in Higher-First Priority:

Scenario_10: 0.9582314936921804

Scenario_9: 0.8038222834527426

Scenario_8: 0.6386992184153006

Scenario_7: 0.4897011323447945

Scenario_6: 0.3610822487815188

Scenario_5: 0.2517311237794721

Scenario_4: 0.15915817873087432

Scenario_3: 0.0901217409995456

Scenario_2: 0.03943721790523324

Scenario_1: 0.011054855384891085

Average MSE across models is 8.933069603449873e-06, indicating strong confidence in the risk assessment.

Follow-up Actions:

- High risk in Scenario_10 (Avg Risk=0.958) driven by metrics: MKC10113, MKC10106, MKR10000. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.
- High risk in Scenario_9 (Avg Risk=0.804) driven by metrics: CPG10107, MSR10107, CPG10102. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.
- High risk in Scenario_8 (Avg Risk=0.639) driven by metrics: CPG10107, MKR10114, SDV10105. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.

Metrics involved in High-Risk Scenarios:

Scenario_10 (Avg Risk=0.958):

- MKC10113: Cost Per Click (CPC)
- MKC10106: Open Rate
- MKR10000: Market Research

Scenario_9 (Avg Risk=0.804):

- CPG10107: Return on Marketing Investment (ROMI)
- MSR10107: Customer Support Response Time
- CPG10102: Impressions

Scenario_8 (Avg Risk=0.639):

- CPG10107: Return on Marketing Investment (ROMI)
- MKR10114: Market Potential
- SDV10105: Customer Acquisition Cost (CAC)

Baseline Predictor (Mean of MKTS_001):

MSE: 1.223451667988889

R-squared: 0.0

RMSE: 1.1060974947936955

MAPE: 1.9635913214388674%

PyTorch Model Evaluation:

MSE: 0.0006558244433992936

R-squared: 0.9994639555770296

RMSE: 0.02560906955356429

MAPE: 0.04542812086958068%

Model Evaluation Metrics:

MSE: 0.0006558244433992936

RMSE: 0.02560906955356429

R²: 0.9994639555770296

MAPE: 0.04542812086958068%

Risk Confidence Score: High

Recommendation - Continuously monitor performance metrics and adjust strategies as needed.

Recommendation - Reassess risk management plans periodically to ensure effectiveness.

-----End of MKTS Module Risk Analysis Summary-----